

Neural-to-Symbolic Evidence Grounding

Progress notes

Working lab notebook, not a paper and not a claim of clinical validity.

17 August 2026 · M1–M3 and Phase 1–4 frozen

Isolated lab: `staging/cxr-evidence-grounding-lab/` UI: <http://127.0.0.1:8253/>

Status of this document. M1–M3, Phase 1 A–D prompts, Phase 2, B', Phase 3, and Phase 4 are frozen. This revision adds the **Phase 4 live record** (second held-out C/U slice, seven notes, three models). Working lab notebook, not a paper and not a claim of clinical validity. Narrative also in `EVALUATION-JOURNEY.md` and `docs/PHASE4-PROTOCOL.md`. Rebuild with `latexmk -pdf progress-notes.tex` from the `notes/` directory.

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1 The question

Can a system reliably determine whether evidence expressed in natural language satisfies a formally defined condition?

Milestone 1 was deliberately tiny: **one** formal condition, **twenty** hand-authored oncology-note snippets, and a **direct-LLM baseline** so the structured pipeline could be compared rather than merely demonstrated.

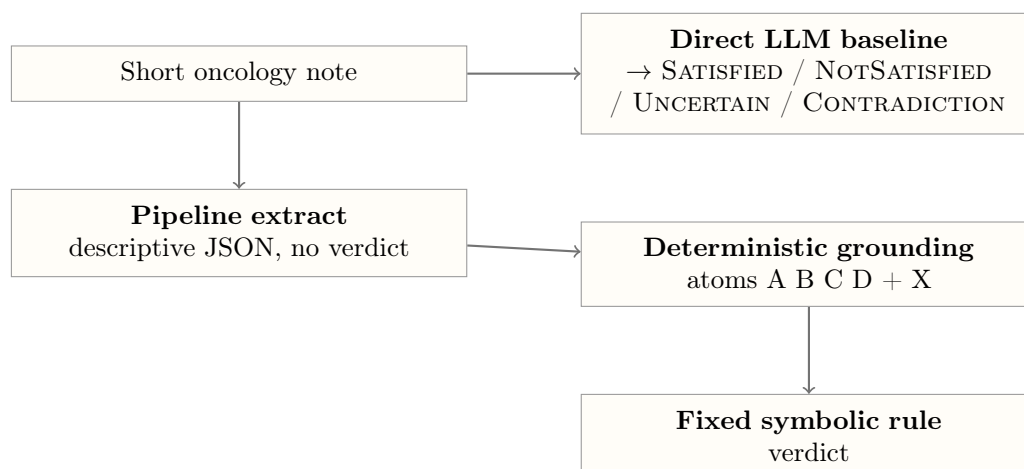
Accuracy was *not* the objective. Version 1 succeeds if the neural-to-symbolic boundary is **observable**—same requirement, different wording, extraction, atoms, rule—and we can mark succeed / fail / uncertain / contradiction.

Hard constraints. The lab does not modify CXR production (`claim_analysis_tools`, Claim Studio, rehearsal UI). It does not add extra predicates, Qdrant, Archetypes, or Next.js. Mock-mode keyword extraction is for inspecting the symbolic path; it is **not** a research result. The direct-LLM baseline is never skipped on live runs. Gold labels are shown in the UI and are **not** sent to the model.

Model. Local Ollama `llama3:8b-instruct-q4_0` at `127.0.0.1:11434`.

2 System under test

Two paths, same notes, same predicate.



Predicate. `FIRST_LINE_THERAPY_FAILED` holds iff all of:

Atom	Name	Meaning
A	first-line identified	A first-line (front-line / 1L) systemic therapy is identified, explicitly or by conventional implication.
B	first-line administered	That therapy was given or attempted, not merely planned.
C	failure event	Progression, non-response, clinician statement of failure/refractory disease, or discontinuation for intolerance without a plan to resume.
D	failure of first-line	The failure event is of that first-line regimen (not only a later line; not completed adjuvant therapy with NED).
X	contradiction	Mutually incompatible claims about A–D.

Rule (unchanged across milestones). `CONTRADICTION` if X; else `NOTSATISFIED` if any of A, B, C, D is false; else `UNCERTAIN` if any is unknown; else `SATISFIED` if $A \wedge B \wedge C \wedge D$.

UI marks. `P+/P-` = pipeline matches / misses **gold**. `B+/B-` = direct LLM matches / misses gold. These are not “the predicate is true.”

3 Evaluation set

Twenty synthetic snippets. Four wording categories, four cases each. Gold is a four-way label, not yes/no. Full text lives in `data/cases.json`. The `why` field is the gold rationale (shown in the UI, not sent to the model).

3.1 Explicit (E) — the note almost says it

ID	Gold	Why included
E1	<code>SATISFIED</code>	Named first-line + explicit progression.
E2	<code>SATISFIED</code>	Stopped for intolerance (counts as line failure).
E3	<code>NOTSATISFIED</code>	Ongoing first-line with response.
E4	<code>NOTSATISFIED</code>	First-line only planned; never given.

3.2 Implicit (I) — you have to infer it

ID	Gold	Why included
I1	<code>SATISFIED</code>	New lesions → switch to second-line (never says “failed”).
I2	<code>SATISFIED</code>	“Platinum-refractory” as conventional 1L failure language.
I3	<code>NOTSATISFIED</code>	Future plan only.
I4	<code>NOTSATISFIED</code>	Completed adjuvant + NED $\neq$ metastatic first-line failure.

3.3 Temporal (T) — order in time matters

ID	Gold	Why included
T1	<code>SATISFIED</code>	Initial response, then later progression on 1L.
T2	<code>NOTSATISFIED</code>	Cycle 1 day 2; too early.
T3	<code>SATISFIED</code>	Response then progression; now on second-line.
T4	<code>NOTSATISFIED</code>	Hold for neutropenia with plan to resume.

3.4 Uncertain (U) — incomplete or mixed

ID	Gold	Why included
U1	UNCERTAIN	Possible progression; awaiting confirmatory scan.
U2	UNCERTAIN	May have received platinum; outcome unknown.
U3	UNCERTAIN	Mixed response.
U4	UNCERTAIN	Unclear whether metastatic 1L or adjuvant; no clear progression.

3.5 Conflicting (C) — two parts of the note disagree

ID	Gold	Why included
C1	CONTRADICTION	Same-day assessment “failed” vs addendum “continue 1L with SD”.
C2	CONTRADICTION	“Never received systemic therapy” vs six cycles then progression.
C3	CONTRADICTION	Referring “failed 1L” vs chart review “never started”.
C4	CONTRADICTION	Same regimen: “progressed” vs later “excellent ongoing response”.

4 Journey of the evaluation

4.1 Milestone 1 — make the boundary visible

Intervention. Isolated lab, one predicate, 20 cases, two paths, UI on port 8253.

Live 20-case run (artifacts/live-m1-before-contradiction.json):

Path	Match gold
Pipeline	13/20
Direct LLM	12/20
Disagree with each other	I2, T2, U1, U3, U4

By category (pipeline / baseline correct): explicit 4/4 and 4/4; implicit 3/4 and 4/4; temporal 4/4 and 3/4; uncertain 2/4 and 1/4; **conflicting 0/4 and 0/4**.

What we learned.

- Explicit cases are easy for both. Structure is not “doing extra work” there.
- **I2:** baseline SATISFIED (correct); pipeline missed at **extraction** (implicit platinum-refractory → B false). The fail is not the rule. Decision: do not chase I2 accuracy.
- **T2, U1, U3:** pipeline closer to gold; baseline over-called SATISFIED or NOTSATISFIED.
- **C1–C4:** both missed all four. On C1 the baseline *described* the addendum conflict, then still emitted NOTSATISFIED.

The interesting failure was contradiction information **recognized in language and dropped before the rule**.

4.2 Milestone 2 — first-class contradiction

Intervention. Extractor must always fill `contradiction.present = true` (two incompatible spans) or false (explicit none). Grounding sets **X** from that field. **Rule unchanged.**

Finding worth writing. The LLM can notice contradiction in prose and still not emit CONTRADICTION as a verdict. Putting two spans on the interface lets the symbolic rule fire.

Live pattern on C (from `artifacts/live-c1-c4.json` and the M2 full run):

Case	Pipeline	Direct LLM
C1	CONTRADICTION	NOTSATISFIED (rationale still describes the conflict)
C2	CONTRADICTION	SATISFIED
C3	CONTRADICTION	NOTSATISFIED
C4	NOTSATISFIED	NOTSATISFIED (present=false)

Headline score stayed about **13/20 vs 12/20**. The *kind* of error changed, not the leaderboard.

Cost of M2. U4 (uncertain line of therapy) was called CONTRADICTION by the pipeline — X over-fired on “metastatic vs adjuvant?”

C4 left failing on purpose. It is not the same *class* as C1–C3 (surface incompatible records / addendum). Progressed-at- T_1 vs excellent-response-at- T_2 on the same regimen might be conflict *or* clinical course. Stretching span extraction to “fix” C4 would hide that class. We do not interpret the miss as proof that a temporal-reasoning module is missing.

4.3 Milestone 2.1 — X is not uncertainty

Intervention. Prompt + grounding: unresolved which-line / mixed / pending → uncertainty, **not** X. Do **not** try to make C4 pass.

Live slice (`artifacts/live-m2.1-c-u4.json`):

Case	Pipeline	Direct LLM	X
C1	CONTRADICTION	NOTSATISFIED	true
C2	CONTRADICTION	SATISFIED	true
C3	CONTRADICTION	NOTSATISFIED	true
C4	NOTSATISFIED	NOTSATISFIED	false (still the hard miss)
U4	NOTSATISFIED	UNCERTAIN	false (no longer CONTRADICTION)

U4 hygiene succeeded (X off). U4 was **not** gold-correct this run (gold UNCERTAIN; pipeline NOTSATISFIED from atoms). That is leftover uncertainty calibration, not a reason to reopen C4.

4.4 Milestone 3 — first-class uncertainty

Intervention. Extractor must always fill `uncertainty.present` = true (one cue + why) or false (explicit none). Grounding keeps implicated A–D atoms **unknown**. **Rule unchanged** (UNCERTAIN still comes from unknown atoms, not a new U atom in the formula). Do not rewrite A–D or X.

Live slice (`artifacts/live-m3-u-c.json`): U1–U4 + C1–C4.

ID	Pipeline	Baseline	U.present	X
U1	UNCERTAIN	SATISFIED	true (pending)	false
U2	UNCERTAIN	NOTSATISFIED	true (unknown outcome)	false
U3	UNCERTAIN	NOTSATISFIED	true	false
U4	UNCERTAIN	UNCERTAIN	true (unresolved line)	false
C1	CONTRADICTION	NOTSATISFIED	false	true
C2	CONTRADICTION	SATISFIED	false	true
C3	CONTRADICTION	NOTSATISFIED	false	true
C4	NOTSATISFIED	NOTSATISFIED	false	false (still miss)

Finding. Same pattern as M2, but **do not treat U1–U4 as one M3 win**. U1/U3 were often already UNCERTAIN from unknown atoms before this field existed. The distinctive M3 save is **U2/U4** no longer collapsing to NOTSATISFIED once **uncertainty.present** sits on the extract→ground contract. The baseline still over-commits on U1–U3 (SATISFIED / NOTSATISFIED); on this run it matched gold on U4. C1–C3 still CONTRADICTION vs baseline not; C4 left in the failure set; U4 X stayed false.

Hypothesis (lab scale; not a new architecture). Reliable neuro-symbolic *verdicts here* depend not only on extracting facts, but on preserving semantic states such as uncertainty and contradiction across the neural-to-symbolic boundary. Scope: one predicate, twenty notes, one local 8B model. The score (about 13/20 vs 12/20) is not the result.

5 What M1–M3 demonstrated (frozen)

The same underlying pattern appeared twice, and the rule engine did not change:

State	Recognized in language	Preserved only when explicit at the N2S boundary
Contradiction	Baseline can <i>describe</i> conflict (C1)	C1–C3: two spans → X → CONTRADICTION
Uncertainty	Baseline can see insufficient evidence	U2/U4: cue+why → unknown atoms → UNCERTAIN

What each slice is for.

- **C1–C3** show that an explicit contradiction state prevents conflict from being flattened into an ordinary verdict.
- **U2/U4** show that an explicit uncertainty state prevents ambiguous evidence from being flattened into SATISFIED or NOTSATISFIED.
- **U1/U3** already often came out UNCERTAIN from unknown atoms; they are supporting cases, not the M3 increment.
- The **direct LLM still over-commits** on U1–U3.
- **C4** remains a deliberately unresolved case: span-pair X is not enough when the apparent conflict depends on regimen identity and time. Gold=CONTRADICTION is a design choice; the miss is the scientific object. This does *not* yet prove that a temporal reasoner is required—only that richer output fields do not cover that class.

Sentence the leftover buys. Explicit state representation solves some boundary failures, but not failures that require deeper temporal or contextual reconciliation. That distinction keeps the result from becoming trivial.

What we are not claiming. Not production CXR. Not that more extract fields will keep fixing errors. Not that the pattern holds beyond this 20-note, one-predicate, one-model lab. A larger controlled set is the *next experiment after this write-up*, not the next field.

6 What we are doing next (Phase 1, in plain language)

M1–M3 are frozen. The next optional experiment does **not** add new medical rules or rebuild CXR.

Everyday picture. Imagine a careful reader who writes notes about a patient chart, then circles one final answer. Sometimes the notes say “these two sentences disagree,” but the circled answer still looks like an ordinary yes/no. We want to know: **at which step did the important idea get lost?**

Four places to look.

1. The written notes never notice the problem.
2. The notes notice it, but the circled answer ignores it.
3. The notes notice it, but our structured checklist drops it.
4. The checklist is right, but our fixed rule still gets the answer wrong.

Four ways we ask the same cases. We reuse only C1–C4 and U1–U4 (same gold labels; we still do not “fix” C4):

- **A** — ask for a final answer directly.
- **B** — ask for free-text notes first, then a final answer from those notes.
- **C** — use the existing structured pipeline (Milestone 3 style).
- **D** — take the free-text notes, convert them into structure, then run the same rule.

What we will report. Not “the score went up.” We will count, when the answer is wrong, **how often each of the four places was the first place meaning disappeared.**

Safe to abandon. This Phase 1 code is optional and separate from Milestones 1–3. If the direction is not useful, delete `run_phase1_diagnostic.py`, `n2s_lab/phase1_diagnostic.py`, and `artifacts/phase1-*.json`. The original lab keeps working.

Technical note (one sentence). Today’s structured LLM calls ask for JSON and parse it; they are not hard-forced field-by-field constrained decoding.

First live Phase 1 numbers (same 8B model; C1–C4 + U1–U4). Match gold: direct answer A = 1/8; notes-then-answer B = 3/8; structured pipeline C = 7/8; notes-then-structure D = 4/8. When the answer missed gold, the first-loss labels were roughly: A — almost all “direct miss”; B — mix of “notes noticed it but answer ignored it” and “notes never noticed it”; C — only C4 still missed (left unresolved on purpose); D — several times the notes had the idea but the

structured form dropped it. These counts are a first diagnostic snapshot, not a claim that the direction is finished.

7 Phase 1 frozen interpretation (path comparison, not “C wins”)

What we freeze. Prompts for Conditions A–D, the eight case ids (C1–C4, U1–U4), gold labels, and the four first-loss stage labels. We do **not** retune prompts after seeing Phase 1 misses, and we leave C4 unresolved by design.

How to read the path comparison. C looking strong on this slice is expected: the structured contract was shaped on the same failure types that appear in these eight notes. That is a **test-set design risk**, not proof that structured extraction is generally best. The scientifically interesting path is often **D**: free-text analysis already carries contradiction or uncertainty, then formalization drops it (**representation_loss**). Keep B’s “analysis→answer mismatch” separate from D’s “analysis→structure loss”—they are different loci at the neural→symbolic boundary.

Defensible claim (lab-scale, pending Phase 2). If the same drop-at-formalization pattern repeats across models and later across unseen wording, we have a research problem around **semantic preservation** at the neural→symbolic interface. If stronger models largely eliminate it, the Phase 1 pattern is more likely an engineering / capability issue for CXR tooling.

8 Phase 2 — cross-model panel (agreed next step)

Protocol. Reuse the frozen A–D prompts and the same eight cases. Rerun on a small local model panel with **no** per-model prompt tuning. Compare whether D’s **representation_loss** (and B’s analysis→answer mismatch) repeats.

Local panel (no external API in this lab run).

- `llama3:8b-instruct-q4_0` — control (M1–M3 / Phase 1 reference); reuse existing Phase 1 artifact when present.
- `mistral:instruct` — different family, similar size.
- `qwen2.5-coder:32b` — stronger local model available on this machine (coder-tuned; still useful as a capacity check).

No higher-precision Llama 8B tag was available locally (`llama3:latest` is also Q4_0). External/API ceiling is deferred unless credentials are provided.

After Phase 2 freezes. Only then: held-out / unseen oncology wording (existing synthetic set if facts are known), still classifying every miss with the same four stages. Do not enlarge *N* for its own sake before the panel is frozen.

Runner. `python3 run_phase2_panel.py` writes `artifacts/phase2-model-panel.json` plus per-model `artifacts/phase2-*.json`.

Live Phase 2 numbers (prompts frozen; same eight cases).

Model	A	B	C	D	D <code>representation_loss</code>
<code>llama3:8b-instruct-q4_0</code> (control)	1/8	3/8	7/8	4/8	3
<code>mistral:instruct</code>	7/8	3/8	7/8	5/8	2
<code>qwen2.5-coder:32b</code>	4/8	7/8	8/8	8/8	0

What this does and does not show. D’s analysis→structure drop is **not Llama-specific**: Mistral still shows two D `representation_loss` errors. It is **not fully general** on this discovery slice: the stronger local 32B model had zero D representation loss and even matched C4 under C and D. B’s analysis→answer mismatch still appears on all three models (3 / 4 / 1 misses). Direct A is highly model-specific (Llama 1/8; Mistral 7/8; Qwen 4/8, missing all four U cases as NOTSATISFIED). This $n = 8$ set is the same wording the contract was tuned on, so it cannot decide the ChatGPT fork (research problem vs capability issue). That decision waits for **unseen** cases with the same frozen prompts and the same four stage labels. Do not retune; do not treat the 32B C4 hit as a reason to change gold.

9 B’ ablation (mapping prompt; frozen B kept)

Why. Phase 2 left analysis→answer mismatch on all three models. A suggested fix was to force the verdict step to “match the scratchpad.” Our analysis prompt forbids categorical labels, so that wording is ill-posed. B’ instead adds an explicit *mapping* rule: incompatible claims → CONTRADICTION; unresolved uncertainty → UNCERTAIN; do not collapse those to SATISFIED/NOTSATISFIED. The free-text analyses are reused from Phase 2; only the verdict call is rerun. Frozen B, A, C, D, and Phase 1/2 JSON are unchanged.

Live B’ numbers (same eight analyses).

Model	B frozen	B’	B mismatch	B’ mismatch
llama3:8b-instruct-q4_0	3/8	3/8	3	3
mistral:instruct	3/8	4/8	4	3
qwen2.5-coder:32b	7/8	8/8	1	0

What moved (and what did not).

- Llama 8B: **no change**. All three analysis→answer mismatches remained (C1, C4, U3). C2/C3 stayed interpretation failures (the prose never clearly named the conflict).
- Mistral: saved **U1 only** (NOTSATISFIED → UNCERTAIN). C1/C3 still collapsed conflict into NOTSATISFIED. C4’s analysis said “conflicting information” and the verdict was still UNCERTAIN, not CONTRADICTION.
- Qwen 32B: finished the last miss (C2 UNCERTAIN → CONTRADICTION). This is the model that was already 7/8.

Discovery (plain language). The B failure on small models is **not** “the prompt forgot to tell the model to read its notes.” We told it, in mapping language, not to flatten conflict or unresolved uncertainty into SATISFIED/NOTSATISFIED. Sub-10B models still flattened **contradiction**. Uncertainty was the easier remap (Mistral U1). Contradiction was the hard one. A one-line consistency patch will not make Llama/Mistral emit CONTRADICTION from conflict prose on this slice.

What we do not conclude. We do **not** conclude that 32B “solved” Condition B or C4. $n = 8$ is the discovery set the contract was tuned on. We do **not** adopt B’ as the Phase 3 default. We do **not** iterate B’ wording further on these eight notes.

What remains open (answered in Phase 3, below). Whether analysis→structure drop (D) and analysis→answer collapse (B) survive **fresh wording** of the same gold types.

Runner. `python3 run_bprime_ablation.py` writes `artifacts/bprime-ablation.json`.

10 Phase 3 — frozen record (held-out C5–C8 / U5–U8)

What we ran. Eight new notes, written as analogs of C1–C4 / U1–U4 with different diseases, drugs, and surface phrasing (`data/heldout-phase3.json`; not merged into the locked 20). Wording was drafted and accepted before any live call. Frozen A–D prompts. B' was **not** used. Same three local models. Same four first-loss labels. C8 gold stayed CONTRADICTION (C4 analog: same named first-line described as progressed and as excellent ongoing response). Phase 1/2 JSON was not overwritten. Runner: `python3 run_phase3_heldout.py` → `artifacts/phase3-heldout-panel.json`.

Headline counts (match gold / 8).

Slice	Model	A	B	C	D	D representation_loss
Discovery (C1–C4, U1–U4)	Llama 8B Q4	1/8	3/8	7/8	4/8	3
	Mistral Instruct	7/8	3/8	7/8	5/8	2
	Qwen 2.5 32B	4/8	7/8	8/8	8/8	0
Held-out (C5–C8, U5–U8)	Llama 8B Q4	1/8	4/8	7/8	6/8	1
	Mistral Instruct	7/8	4/8	7/8	6/8	2
	Qwen 2.5 32B	3/8	6/8	8/8	6/8	1

What happened, case by case (held-out). Gold: C5–C8 = CONTRADICTION; U5–U8 = UNCERTAIN. Cells are the four condition verdicts (A / B / C / D).

ID	Analog	Llama 8B	Mis
C5	C1 same-day reverse	NOTSATISFIED / CONTRADICTION / CONTRADICTION / CONTRADICTION	Co
C6	C2 never vs cycles	NOTSATISFIED / UNCERTAIN / CONTRADICTION / UNCERTAIN	Co
C7	C3 referred fail vs never given	NOTSATISFIED / UNCERTAIN / CONTRADICTION / CONTRADICTION	Co
C8	C4 same regimen both ways	NOTSATISFIED / SATISFIED / NOTSATISFIED / NOTSATISFIED	UN
U5	U1 pending scan	NOTSATISFIED / UNCERTAIN / UNCERTAIN / UNCERTAIN	UN
U6	U2 unknown outside	NOTSATISFIED / UNCERTAIN / UNCERTAIN / UNCERTAIN	UN
U7	U3 mixed	NOTSATISFIED / NOTSATISFIED / UNCERTAIN / UNCERTAIN	UN
U8	U4 which-line	UNCERTAIN / UNCERTAIN / UNCERTAIN / UNCERTAIN	UN

Named losses (when verdict ≠ gold).

- **D representation_loss (analysis had the distinction; structure dropped it):** Llama C8; Mistral C6 and C8; Qwen C6. Qwen had **zero** of these on the discovery eight; C6 is the counterexample on unseen wording.
- **D symbolic_failure (structure preserved; rule still missed):** Qwen C7 only (D emitted UNCERTAIN after C was CONTRADICTION).
- **B analysis→answer mismatch:** Llama C7, C8, U7; Mistral C5, C6, C7, C8; Qwen C6, C8. Present on all three models on held-out notes.
- **C miss:** Llama and Mistral only on **C8** (C4 analog). Qwen C was 8/8.
- **Direct A:** Llama 1/8 (only U8); Mistral 7/8 (miss C8); Qwen 3/8 (all four U as NOTSATISFIED, plus C6).

Frozen reading (lab-scale). On wording the prompts were not designed around, the same two loci still appear: (1) free-text analysis carries conflict or uncertainty, then the categorical answer ignores it (B); (2) free-text analysis carries the distinction, then structured extraction drops it (D). (2) is not Llama-specific and is not confined to the discovery eight: the 32B model that looked clean on that slice dropped C6 at formalization. C remaining strong on this slice is still partly contract-shaped (same gold types); C8/C4 remains the identity/time miss and gold stays CONTRADICTION. B' is not the Phase 3 default and was not used.

What this does not conclude. Not a paper result. Not clinical validity. Not “C wins.” Not that 32B is useless or that small models cannot be engineered around. $n = 8$ held-out is still small. No API ceiling was run. We did not retune gold or prompts after seeing these misses.

Next protocol. Phase 4 (below) was the planned replication. No further N2S protocol is queued unless designed separately.

11 Phase 4 protocol (frozen)

Question. When NL analysis already has contradiction or uncertainty, how often does that distinction survive formalization (D) and categorical answer (B), on wording the prompts were not tuned on? Not whether C is “more accurate.” Not a new failure class.

Object. Two **separate** conditional loss rates among cases where a **human** marks that the free-text analysis states the gold distinction: B loss = analysis had it, Condition B verdict $\neq$ gold; D loss = analysis had it, Condition D is **representation_loss**. Regex cue flags are a screen only, not the headline denominator. Match-gold/ n is secondary.

Controls. Frozen A–D prompts. B' off. Temperature 0 (no resampling the same note). Gold not fed to models; gold not changed after output. Do not overwrite Phase 1–3 artifacts.

Data. Seven notes, one analog each of C1, C2, C3, U1, U2, U3, U4. Ids C9–C11 and U9–U12 in `data/heldout-phase4.json`. **No C4/C8 analog** (identity/time waits). Wording accepted 17 August 2026; live A–D run same day. Not merged into `data/cases.json`.

Models. `llama3:8b-instruct-q4_0`, `mistral:instruct`, `qwen2.5-coder:32b`. No API. No per-model prompts.

Falsification (declared before run). If D loss = 0 on all three models among human-confirmed analyses, the formalization-loss claim is weakened. If D loss recurs on more than one model, that claim gains support. B-only vs D-only findings stay separate. C8-class misses are out of scope.

Not in Phase 4. Iterate B'; fix C8; extra extract fields; CXR production; grow N for its own sake; peek at model output while drafting notes.

Durable copy. `docs/PHASE4-PROTOCOL.md`.

12 Phase 4 — frozen record (held-out C9–C11 / U9–U12)

What we ran. Seven new notes, written as analogs of C1–C3 / U1–U4 only (`data/heldout-phase4.json`; not merged into the locked 20). **No C4/C8 analog.** Wording was drafted and accepted before any live call. Frozen A–D prompts. B' was **not** used. Same three local models. Same four first-loss labels. Phase 1/2/3 JSON was not overwritten. Runner: `python3 run_phase4.py` → `artifacts/phase4-heldout-panel.json`.

Headline counts (match gold / 7).

Slice	Model	A	B	C	D	D representation_loss
Held-out (C9–C11, U9–U12)	Llama 8B Q4	2/7	4/7	6/7	5/7	1
	Mistral Instruct	7/7	4/7	7/7	4/7	1
	Qwen 2.5 32B	2/7	5/7	6/7	6/7	1

What happened, case by case (held-out). Gold: C9–C11 = CONTRADICTION; U9–U12 = UNCERTAIN. Cells are the four condition verdicts (A / B / C / D).

ID	Analog	Llama 8B	Mistral
C9	C1 same-day reverse	NOTSATISFIED / UNCERTAIN / CONTRADICTION / CONTRADICTION	CONTR
C10	C2 never vs cycles	SATISFIED / NOTSATISFIED / SATISFIED / UNCERTAIN	CONTR
C11	C3 referred fail vs never given	SATISFIED / NOTSATISFIED / CONTRADICTION / UNCERTAIN	CONTR
U9	U1 pending scan	NOTSATISFIED / UNCERTAIN / UNCERTAIN / UNCERTAIN	UNCER
U10	U2 unknown outside	NOTSATISFIED / UNCERTAIN / UNCERTAIN / UNCERTAIN	UNCER
U11	U3 mixed / deferred	UNCERTAIN / UNCERTAIN / UNCERTAIN / UNCERTAIN	UNCER
U12	U4 which-line	UNCERTAIN / UNCERTAIN / UNCERTAIN / UNCERTAIN	UNCER

Named losses (when verdict ≠ gold).

- **D representation_loss (analysis had the distinction; structure dropped it):** Llama C11; Mistral C11; Qwen C10. **All three models** had exactly one D-rep-loss on this slice (declared stop rule: formalization-loss claim **gains support**).
- **B analysis→answer mismatch:** Llama C9, C10, C11; Mistral C9, C10, C11; Qwen C9, C10. Present on all three models.
- **C miss:** Llama and Qwen on **C10** (C2 analog: never vs cycles). Mistral C was 7/7.
- **Direct A:** Llama 2/7 (U11, U12); Mistral 7/7; Qwen 2/7 (all four U as NOTSATISFIED, same pattern as Phase 3).
- **U notes:** all four U matched B, C, and D on all three models.

Regex screen (not headline). Among rows where `analysis_has_distinction` fired on the B or D analysis text: D `representation_loss` counts were 1/6 (Llama), 1/5 (Mistral), 1/7 (Qwen). The protocol headline still requires human confirmation (A_{human}); regex ≠ “analysis is correct.”

Frozen reading (lab-scale). Phase 4 was the sharper replication ask: when analysis already carries conflict or uncertainty, does that survive categorical answer (B) and formalization (D) on a **second** independent held-out slice? On this seven-note set, both loci recurred: (1) B mismatch on all three models on all three C notes; (2) D `representation_loss` on all three models (one case each: C11/C11/C10). Uncertainty notes (U9–U12) did not show B or D loss on any model.

This strengthens the lab-scale formalization-loss reading from Phase 1–3; it is still not a paper claim and $n = 7$ is small.

What this does not conclude. Not clinical validity. Not that C-path “solved” preservation (C10 miss on two models). Not that human A_{human} rates are final without reading the analyses. No API ceiling. Gold and prompts were not retuned after seeing misses. No further N2S run is queued unless a new protocol is designed.

Runner. `python3 run_phase4.py` writes `artifacts/phase4-heldout-panel.json` and per-model `phase4-*.json`.

13 What we are evaluating (and what we are not)

Evaluating.

- Whether a direct LLM verdict and a structured extract→ground→rule path **differ** on the same 20 statements.
- Whether an explicit field at the neural→symbolic boundary **preserves** a semantic state (contradiction) that the baseline narrates then discards.
- Whether that field **over-generalizes** (U4) and can be constrained without “fixing” a harder class (C4).
- Whether the same move for **uncertainty** (`uncertainty.present` → unknown atoms) stops collapse to NOTSATISFIED while the baseline still over-commits.
- Whether a mapping constraint at Condition B (B′) removes analysis→answer mismatch on the frozen eight analyses (it did not, for sub-10B contradiction).
- Whether B mismatch and D **representation_loss** survive held-out C/U wording under frozen A–D (Phase 3: both did, including 32B D-loss on C6; Phase 4: both again on a second seven-note slice, D-rep-loss on all three models).

Not evaluating (yet).

- Production CXR / Archetypes / Qdrant / kernel fusion.
- Accuracy as a goal, or a larger gold set.
- Additional predicates.
- Whether C4 *should* be CONTRADICTION vs course-over-time (gold is a design choice; the miss is the scientific object).

14 How to read a case in the UI

1. Open <http://127.0.0.1:8253/> → **Load last artifact** (or **Run selected**).
2. Left list: category + P+ B+ vs gold.
3. Right pane, top to bottom: evidence + gold → baseline vs pipeline verdicts → neural JSON → **Contradiction field (X)** → **Uncertainty field** → grounded atoms A–D + X → the **same** predicate text on every case.

For the write-up, the instructive cases are **C1** (structure preserves conflict), **U2/U4** (uncertainty no longer collapses to NOTSATISFIED), **U4 vs X** (uncertainty must not be labeled contradiction), and **C4** (harder identity/time class, left failing).

15 Artifact index

File	What it is
data/cases.json	Locked 20 cases + gold + why.
data/heldout-phase3.json	Frozen held-out C5–C8 / U5–U8 (not merged into the locked 20).
data/heldout-phase4.json	Frozen held-out C9–C11 / U9–U12 (not merged into the locked 20).
artifacts/phase3-heldout-panel.json	Phase-3 cross-model summary (held-out A–D).
artifacts/phase3-*.json	Per-model Phase-3 A–D rows.
run_phase3_heldout.py	Phase-3 held-out panel runner (frozen A–D; no B').
artifacts/phase4-heldout-panel.json	Phase-4 cross-model summary (held-out A–D).
artifacts/phase4-*.json	Per-model Phase-4 A–D rows.
run_phase4.py	Phase-4 held-out panel runner (frozen A–D; no B').
artifacts/live-m1-before-contradiction.json	M1 full live 20.
artifacts/live-c1-c4.json	M2 fresh C1–C4.
artifacts/live-m2.1-c-u4.json	M2.1 C1–C4 + U4.
artifacts/live-m3-u-c.json	M3 U1–U4 + C1–C4.
artifacts/phase1-transition-c-u.json	Optional Phase-1 A–D diagnostic (removable).
artifacts/phase2-model-panel.json	Phase-2 cross-model summary (same A–D / 8 cases).
artifacts/phase2-*.json	Per-model Phase-2 full A–D rows.
artifacts/bprime-ablation.json	B' mapping-prompt ablation vs frozen B.
run_bprime_ablation.py	B' runner (does not overwrite Phase-1/2).
docs/PHASE1-DIAGNOSTIC.md	Plain how-to + undo for Phase-1.
docs/PHASE4-PROTOCOL.md	Frozen Phase-4 protocol + live record in this PDF.
run_phase2_panel.py	Frozen-prompt model panel runner.
artifacts/live-latest.json	Last viewer cache (U/C rows merged from M3; E/I/T may predate M3).
artifacts/mock-*.json	Heuristic mock; not a result.
EVALUATION-JOURNEY.md	Durable markdown narrative if JSON is missing.
notes/progress-notes.pdf	This document.

Live JSON is gitignored. The markdown journey note and this PDF are the durable narrative.

16 One-paragraph abstract

We built a 20-note lab that asks whether natural-language oncology snippets satisfy a single formal predicate, `FIRST_LINE_THERAPY_FAILED`, comparing a direct local LLM verdict to an extract→ground→rule pipeline. Explicit wording is easy for both. The first scientific result is not a higher score (13/20 vs 12/20) but a failure mode: the LLM can describe a contradiction and still not emit `CONTRADICTION`. Making contradiction a required extract field (two spans or none) let the unchanged symbolic rule fire on C1–C3 while the baseline still would not. C4 remains unresolved because the apparent conflict depends on regimen identity and time, not a surface span pair. A follow-up constraint stopped the new field from labeling unresolved

line-of-therapy (U4) as contradiction. The same move for uncertainty (`uncertainty.present` → keep atoms unknown) stopped **U2/U4** collapsing to NOTSATISFIED while the direct LLM still over-committed on U1–U3. U1/U3 were often already pipeline-UNCERTAIN from unknown atoms; they are not the M3 increment. C4 remains the harder identity/time miss—explicit state fields do not cover that class. This is not production CXR.

Phase 1–4 addendum (lab-scale). Conditions A–D on discovery C1–C4/U1–U4, held-out C5–C8/U5–U8, then a second held-out C9–C11/U9–U12 ($n = 7$, no C8 analog), three local models, prompts frozen. The 32B model had zero D `representation_loss` on the discovery eight and one each on Phase 3 (C6) and Phase 4 (C10). B analysis→answer mismatch appeared on all three models on all three slices. B' mapping did not fix sub-10B contradiction collapse. Phase 4 declared stop rule: D-rep-loss on all three models → formalization-loss reading gains support (still lab-scale, not a paper). C8/C4 analog remains the identity/time miss; gold stays CONTRADICTION. No further N2S protocol is queued unless designed separately.

A Note texts (locked set)

The following are the twenty snippets as stored in `data/cases.json` on 17 August 2026.

ID	Gold	Evidence
E1	SATISFIED	Metastatic NSCLC. Patient progressed on first-line carboplatin and paclitaxel after four cycles. Plan to start second-line immunotherapy.
E2	SATISFIED	First-line cisplatin/gemcitabine was stopped after two cycles due to grade 3 neuropathy. Therapy discontinued for intolerance; switching regimens.
E3	NOTSATISFIED	Currently receiving first-line pembrolizumab plus pemetrexed. Restaging CT shows ongoing partial response. Continue current therapy.
E4	NOTSATISFIED	New diagnosis of metastatic pancreatic adenocarcinoma. No prior systemic therapy. Starting first-line FOLFIRINOX next week.
I1	SATISFIED	After carboplatin/paclitaxel, imaging showed new liver lesions. We are switching to second-line nivolumab.
I2	SATISFIED	Platinum-refractory high-grade serous ovarian cancer. Now considering pegylated liposomal doxorubicin.
I3	NOTSATISFIED	Medical oncology visit: plan to start carboplatin and paclitaxel next week after port placement. ECOG 1. Staging complete.
I4	NOTSATISFIED	History of stage II colon cancer. Patient completed adjuvant FOLFOX two years ago and remains with no evidence of disease on surveillance.
T1	SATISFIED	Started first-line carboplatin/pemetrexed in January 2025 with initial radiographic response. CT in August 2025 showed progression in the mediastinum.
T2	NOTSATISFIED	First-line FOLFOX was initiated yesterday. No restaging yet. Patient tolerating cycle 1 day 2.
T3	SATISFIED	Responded to first-line carboplatin/etoposide for eight months, then progressed. Currently on second-line topotecan.
T4	NOTSATISFIED	Cycle 3 of first-line carboplatin/paclitaxel held last week for neutropenia. Plan to resume at reduced dose. No evidence of progression.

ID	Gold	Evidence
U1	UNCERTAIN	Possible progression on first-line osimertinib; awaiting confirmatory scan next week before changing therapy.
U2	UNCERTAIN	Outside records suggest the patient may have received platinum-based therapy at another facility. Outcome of that treatment is unknown. No source documents available.
U3	UNCERTAIN	On first-line FOLFIRINOX. Mixed response: liver lesions smaller, new questionable peritoneal nodularity. Oncology to discuss next steps after tumor board.
U4	UNCERTAIN	Questionable whether carboplatin was given as first-line metastatic therapy or in the adjuvant setting. No clear progression documented in the available notes.
C1	CONTRADICTION	Assessment: first-line therapy failed, proceed to second-line. Addendum later the same day: patient continues first-line carboplatin with stable disease; do not change therapy.
C2	CONTRADICTION	Intake note: never received systemic therapy. Later oncology note: progressed after six cycles of first-line carboplatin/paclitaxel.
C3	CONTRADICTION	Failed first-line per referring oncologist. Our chart review: first-line systemic therapy was never started; patient received palliative radiotherapy only.
C4	CONTRADICTION	Progressed on carboplatin/paclitaxel (first-line). Later note: excellent ongoing response to first-line carboplatin/paclitaxel; continue current treatment.

B Held-out note texts (Phase 3, frozen)

The following are the eight snippets as stored in `data/heldout-phase3.json` on 17 August 2026. They were not in the prompt-design set.

ID	Gold	Evidence
C5	CONTRADICTION	AM oncology: first-line FOLFOX has failed; start second-line FOLFIRI this week. PM addendum the same calendar day: CT today shows stable disease on first-line FOLFOX; continue current regimen, no switch.
C6	CONTRADICTION	ED triage: patient reports never having chemotherapy. Medical oncology follow-up: completed six cycles of first-line gemcitabine/nab-paclitaxel, then radiographic progression in the liver.
C7	CONTRADICTION	Outside consultation letter: failed first-line systemic therapy, referred for next-line options. Internal chart review: no first-line systemic therapy was administered here or elsewhere; treatment to date is best supportive care only.
C8	CONTRADICTION	Interval note: progression on first-line palbociclib plus letrozole. Subsequent clinic note: excellent ongoing response to first-line palbociclib plus letrozole; continue current treatment.
U5	UNCERTAIN	Possible progression on first-line sunitinib; MRI pending next Thursday before any regimen change.
U6	UNCERTAIN	Family states the patient may have received immunotherapy at an outside clinic last year. No records transferred. Whether that treatment was given, and how it went, is unknown.

ID	Gold	Evidence
U7	UNCERTAIN	On first-line R-CHOP. Mixed response: mediastinal nodes smaller, new questionable splenic lesions. Plan deferred until lymphoma conference.
U8	UNCERTAIN	Unclear from the scanned notes whether oxaliplatin was given as first-line therapy for metastatic disease or as adjuvant treatment after resection. No definite progression is documented.

C Held-out note texts (Phase 4, frozen)

The following are the seven snippets as stored in `data/heldout-phase4.json` on 17 August 2026. No C4/C8 analog. They were not in the prompt-design set.

ID	Gold	Evidence
C9	CONTRADICTION	07:40 tumor-board list: first-line ipilimumab/nivolumab has failed; schedule second-line BRAF-directed therapy. 19:10 same-day clinic addendum: restaging today is stable disease on first-line ipilimumab/nivolumab; keep current immunotherapy, do not switch.
C10	CONTRADICTION	Pre-op H&P: no prior cytotoxic chemotherapy. Lymphoma clinic later: finished four cycles of first-line ABVD, then PET showed progressive nodal disease.
C11	CONTRADICTION	Referring packet: failed first-line systemic therapy, sent for next-line trial screening. Pharmacy and MAR review here: no first-line systemic anticancer drug was ever dispensed; care to date is hospice and pain control only.
U9	UNCERTAIN	Possible radiographic progression on first-line enzalutamide. Bone scan results are not back; no regimen change until they return.
U10	UNCERTAIN	Patient mentions maybe receiving docetaxel at a VA facility. Those records have not arrived. We do not know if that therapy was given or what happened.
U11	UNCERTAIN	On first-line pazopanib. Mixed picture: target lesions smaller, new questionable peritoneal nodules. No change until sarcoma board next week.
U12	UNCERTAIN	Cannot tell from the scanned summaries whether doxorubicin was given as first-line therapy for metastatic breast cancer or as adjuvant treatment after mastectomy. No definite progression is recorded.